

The wedge product of the coordinate differentials dq_i, dp_i may be written

$$dq_i \wedge dp_i(\boldsymbol{\xi}, \boldsymbol{\eta}) = \xi_i \eta_{i+N_c} - \xi_{i+N_c} \eta_i = \boldsymbol{\xi}^T \mathbf{J}^{(i)} \boldsymbol{\eta},$$

where $\mathbf{J}^{(i)}$ is the matrix which has zeros everywhere except for $(\mathbf{J}^{(i)})_{i,i+N_c} = 1$, $(\mathbf{J}^{(i)})_{i+N_c,i} = -1$. Summing these terms results in the *symplectic 2-form*, denoted ψ_S :

$$\psi_S = \sum_{i=1}^{N_c} dq_i \wedge dp_i(\boldsymbol{\xi}, \boldsymbol{\eta}) = \boldsymbol{\xi}^T \left(\sum_{i=1}^{N_c} \mathbf{J}^{(i)} \right) \boldsymbol{\eta} = \boldsymbol{\xi}^T \mathbf{J} \boldsymbol{\eta}.$$

Thus the matrix

$$\mathbf{J} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & \mathbf{0} \end{bmatrix}$$

defines a representation for the 2-form ψ_S . It is easy to verify that

$$\psi_S = \sum_{i < j} dz_i \wedge J_{ij} dz_j.$$

In general, a differential 2-form ψ is represented in coordinates by

$$\psi_z = \sum_{i,j} a_{ij}(\mathbf{z}) dz_i \wedge dz_j,$$

with matrix of coefficients $\mathbf{A}(\mathbf{z}) = (a_{ij}(\mathbf{z}))$.

The *pull-back* of a differential 1-form ψ under a phase space mapping Φ is defined as the action of ψ after transformation by the Jacobian matrix of the mapping. It is written $\Phi^* \psi$, so

$$(\Phi^* \psi_z)(\boldsymbol{\xi}) = \psi_{\Phi(\mathbf{z})}(\Phi'(\mathbf{z})\boldsymbol{\xi}).$$

The pull-back of a differential 2-form $\psi_1 \wedge \psi_2$ is consequently defined as

$$\Phi^*(\psi_1 \wedge \psi_2) = (\Phi^* \psi_1) \wedge (\Phi^* \psi_2).$$

Given a differential 2-form ψ_z represented by the matrix $\mathbf{A}(\mathbf{z}) = (a_{ij}(\mathbf{z}))$, the *pull-back* of ψ_z under Φ is defined by

$$\Phi^* \psi_z = \sum_{i,j} b_{ij}(\mathbf{z}) dz_i \wedge dz_j,$$

where the matrix $\mathbf{B}(\mathbf{z}) = (b_{ij}(\mathbf{z}))$ is related to $\mathbf{A}(\mathbf{z})$ by

$$\mathbf{B}(\mathbf{z}) = \Phi'^T(\mathbf{z})\mathbf{A}(\Phi(\mathbf{z}))\Phi'(\mathbf{z}).$$

We say that a 2-form ψ is *conserved under mapping* Φ if

$$\Phi^*\psi = \psi.$$

In coordinates, the conservation of the 2-form represented by matrix $\mathbf{A}$ under a mapping Φ means that

$$\Phi'^T \mathbf{A} \Phi' = \mathbf{A}.$$

In the particular case of the symplectic 2-form ψ_s , we have $\mathbf{A} = \mathbf{J}$, and the following condition for conservation under the mapping Φ

$$\Phi'^T \mathbf{J} \Phi' = \mathbf{J}. \quad (2.17)$$

A map that conserves the symplectic 2-form, or, in coordinates, satisfies (2.17), is termed a *symplectic map*.

Taking the determinant of both sides of (2.17), we have

$$\det(\Phi'^T \mathbf{J} \Phi') = \det(\mathbf{J}) \Rightarrow \det(\Phi'^T) \det(\mathbf{J}) \det(\Phi') = \det(\mathbf{J}),$$

hence

$$\det(\Phi')^2 = 1,$$

so $|\det(\Phi')| = 1$. This raises a curious issue since we could conceivably have $\det(\Phi') = -1$.

In the case of a flow map $\Phi = \mathcal{F}_t$, we know that for $t \rightarrow 0$, the map reduces to the identity map (the same has to happen for a consistent numerical method). This means that we would have to have

$$\lim_{t \rightarrow 0} \det(\mathcal{F}'_t) = 1.$$

If the system is Hamiltonian, the map is symplectic for all t , and the determinant will be a continuous function of t , so the cases of interest have $\det(\mathcal{F}'_t) = +1$. The flow map of a Hamiltonian system is volume preserving.